Skooly System Design Analysis

1. Project Overview

Skooly is a multi-tenant school management SaaS platform developed as part of my thesis work. Unlike traditional single-school systems, Skooly is designed as a **networked platform** where schools, teachers, and families can connect across the ecosystem. The platform aims to provide a single, unified application for managing academic operations, people, scheduling, and billing—with a roadmap toward cross-school features such as teacher mobility and student transfer and inter-school activities.

1.1 Core Capabilities

- **Five user roles**: Admin, Teacher, Student, Parent, System Admin — all in one unified app with role-based views
- **Academic operations**: Academic year management, classes, grades, subjects, curriculum, scheduling, lessons, exams, assignments, attendance, results
- **People management**
- **Subscription billing**: Stripe integration for plan-based pricing per school

1.2 What Differentiates Skooly

Aspect	Skooly
Model	Multi-tenant SaaS, subscription-based (Stripe)
Teacher mobility	Teacher marketplace (planned): teachers can work at multiple schools, discoverable by schools
Student mobility	Student transfer between schools (planned); parent access across children's schools
Curriculum	Optional country-based templates (France, USA, UK, Morocco, IB/Cambridge)
Schedule changes	Teacher-initiated requests (time change, swap); admin approves with conflict checks
Cross-school	Roadmap: sports leagues, alumni network, group purchasing, shared transport
Tech stack	Next.js 14, Prisma, PostgreSQL, Server Actions, App Router

1.3 Tech Stack

- **Frontend/Backend**: Next.js 14 (App Router, Server Actions, RSC)
- **Database**: PostgreSQL via Prisma ORM
- **Auth**: Custom JWT (HttpOnly cookie), no third-party auth
- **Payments**: Stripe (subscriptions)
- **Media**: Cloudinary (images)

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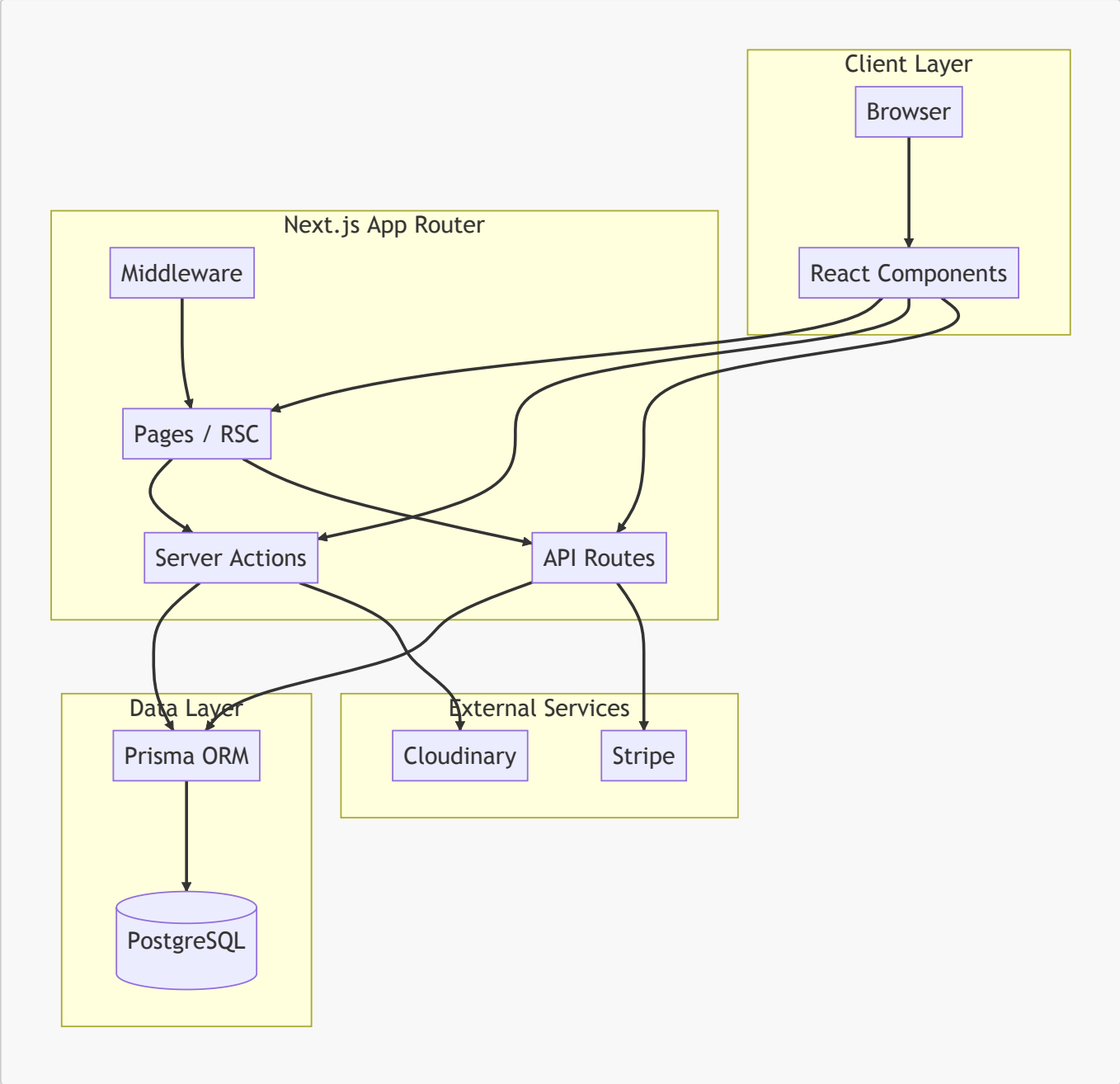
2. High-Level Architecture



Key architectural decisions:

- **Hybrid data access**: Server Actions (primary for CRUD) + REST API routes (auth, webhooks, subscriptions)
- **JWT-based auth**: Custom auth with `auth_token` HttpOnly cookie, no Clerk (removed)
- **Multi-tenant**: School-scoped data via `schoolId` on all entities
- **Middleware**: Route protection, role-based redirects, school isolation

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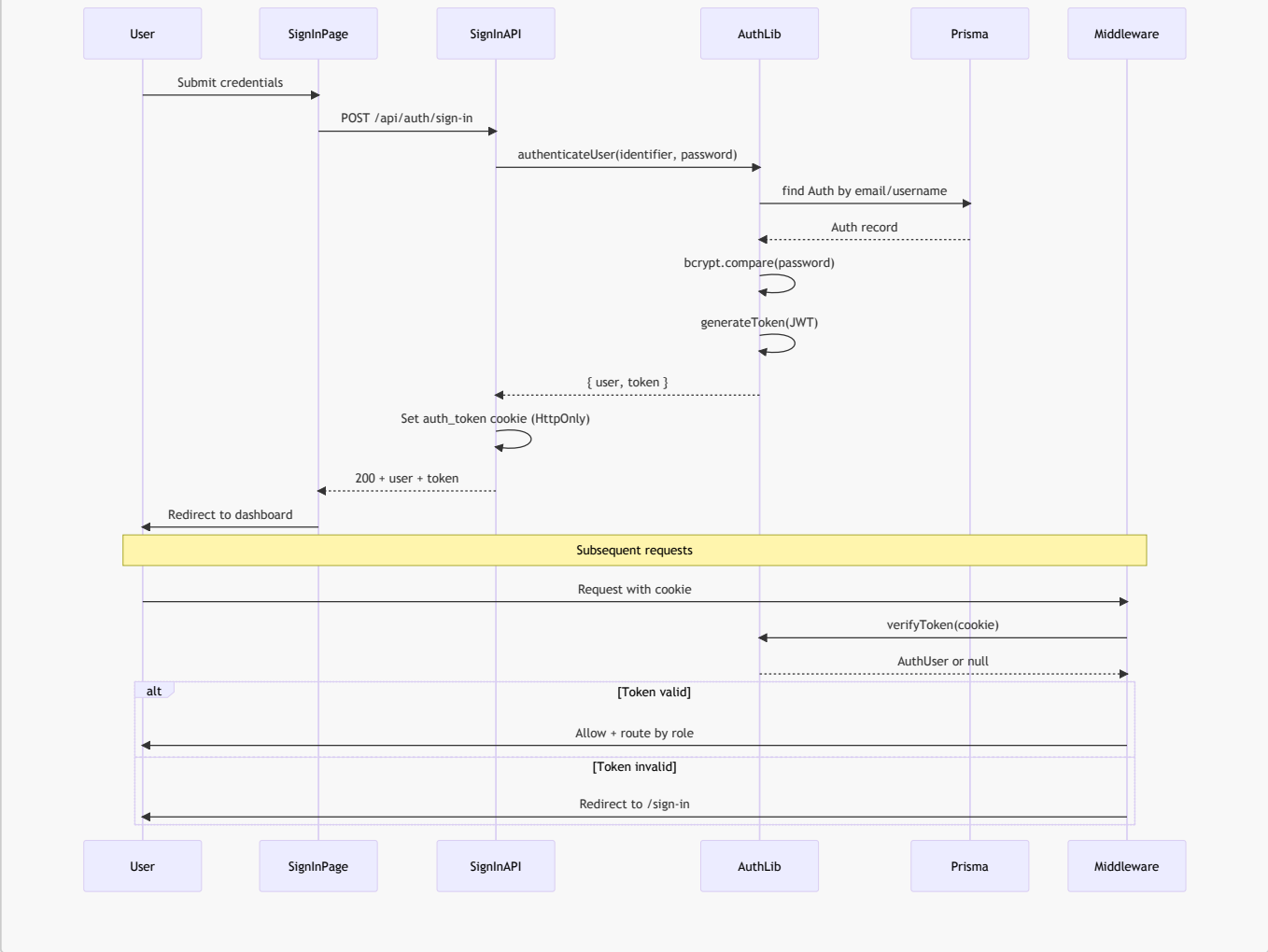
4. Authentication Flow



Auth components:

- src/lib/auth.ts: `authenticateUser`, `verifyToken`, `requireAuth`, `requireRole`, `requireSchoolAccess`, `getCurrentUserOnPage`
- src/middleware.ts: Route protection, role-based redirects (`/schools/[schoolId]/[role]`, `/system/*` for system_admin)
- Token: JWT (HS256, 7d expiry), stored in `auth_token` HttpOnly cookie

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5. User Onboarding Flows

5.2 School Admin (New School)

Who: Person creating a new school on the platform.

Flow: Single-step account + school creation.

Step	Action	Location
1	User clicks "Get Started"	`/` or `/create-school`
2	Form: email, password, confirmPassword, schoolName	`/create-school`
3	Server Action: `createSchoolAndAssignAdmin`	Creates Auth + School + Admin in one transaction

Step	Action	Location
4	Set cookie via `/api/auth/set-token` (new user) or redirect (existing user)	Client
5	Redirect to `/schools/[schoolId]/admin`	Dashboard

Existing user variant: If already logged in (no school), form shows schoolName only; user is linked to new school as admin.

5.3 Existing User (Sign In)

Who: Admin, Teacher, Student, Parent, or System Admin with an existing account.

Flow:

Step	Action	Location
1	User enters email/username + password	`/sign-in`
2	POST `/api/auth/sign-in`	Sets `auth_token` cookie
3	Redirect by role	
	system_admin	`/system/plans`
	admin	`/schools/[schoolId]/admin`
	teacher	`/schools/[schoolId]/list/teachers/[profileId]` or `/teacher`
	student	`/schools/[schoolId]/list/students/[profileId]` or `/student`
	parent	`/schools/[schoolId]/parent`
4	If no schoolId	Redirect to `/create-school`

5.4 Invited User (Teacher, Student, Parent)

Who: Invited by admin to join an existing school.

Flow:

Step	Action	Location
1	Admin sends invite (email, role, optional metadata)	Admin UI or bulk CSV
2	System creates Invite record, sends email with link	`/invite/[token]`
3	User clicks link	`/invite/[token]`
4	System validates token (exists, PENDING, not expired)	Server
5	User sets password (+ optional profile fields)	Accept form

Step	Action	Location
6	System creates Auth + profile (Teacher/Student/Parent), marks invite ACCEPTED	Server Action
7	Redirect to dashboard	`/schools/[schoolId]/[role]`

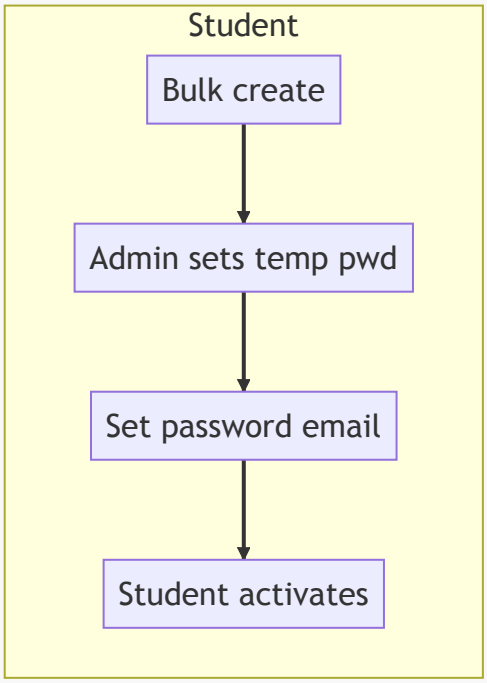
Admin side: See Section 6 (People Management) for sending invites. This section describes the invited user's flow.

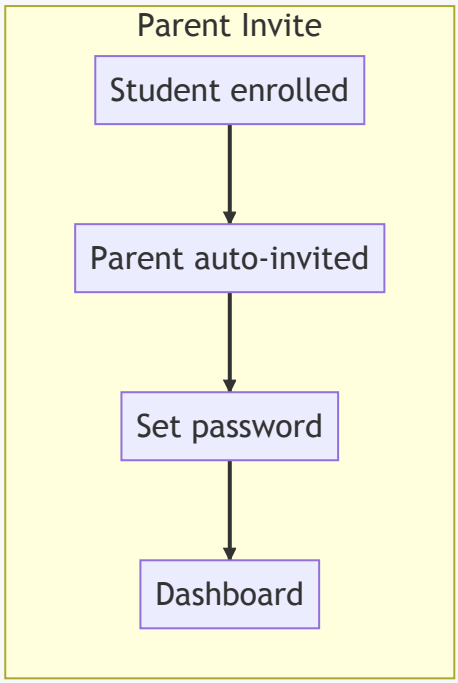
5.5 Bulk Onboarding (Scale)

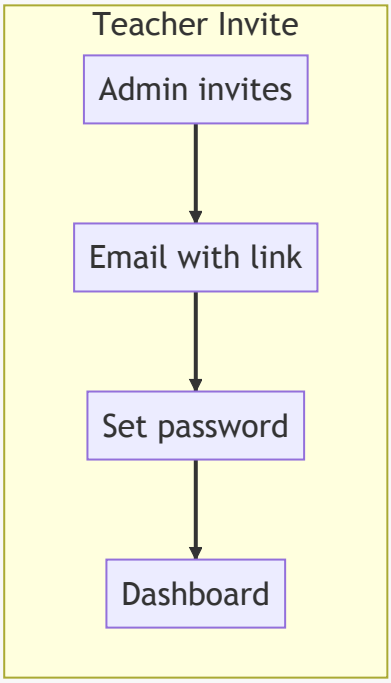
For schools with many users, admin uses bulk operations:

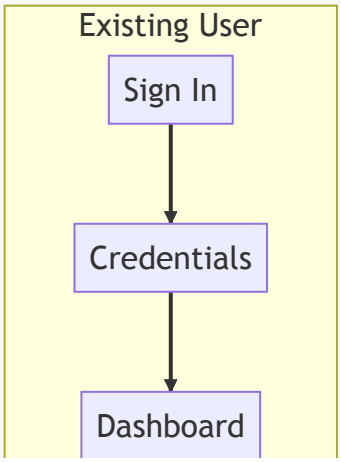
Role	Method	How
Teachers	Bulk invite	CSV upload: email, role, name, subjects → create Invites, send emails
Teachers	Bulk create + invite	Extend bulk import: create profiles, send "Set your password" emails
Parents	Invite on enrollment	When admin creates/enrolls student, auto-invite parent (email from student record)
Parents	Bulk invite	CSV: email, studentIds → create Invites
Students	Bulk create	Admin bulk creates; send "Set password" email to parent (or student if email exists)
Students	Parent-managed	Parent creates account first; parent sets up student login when needed

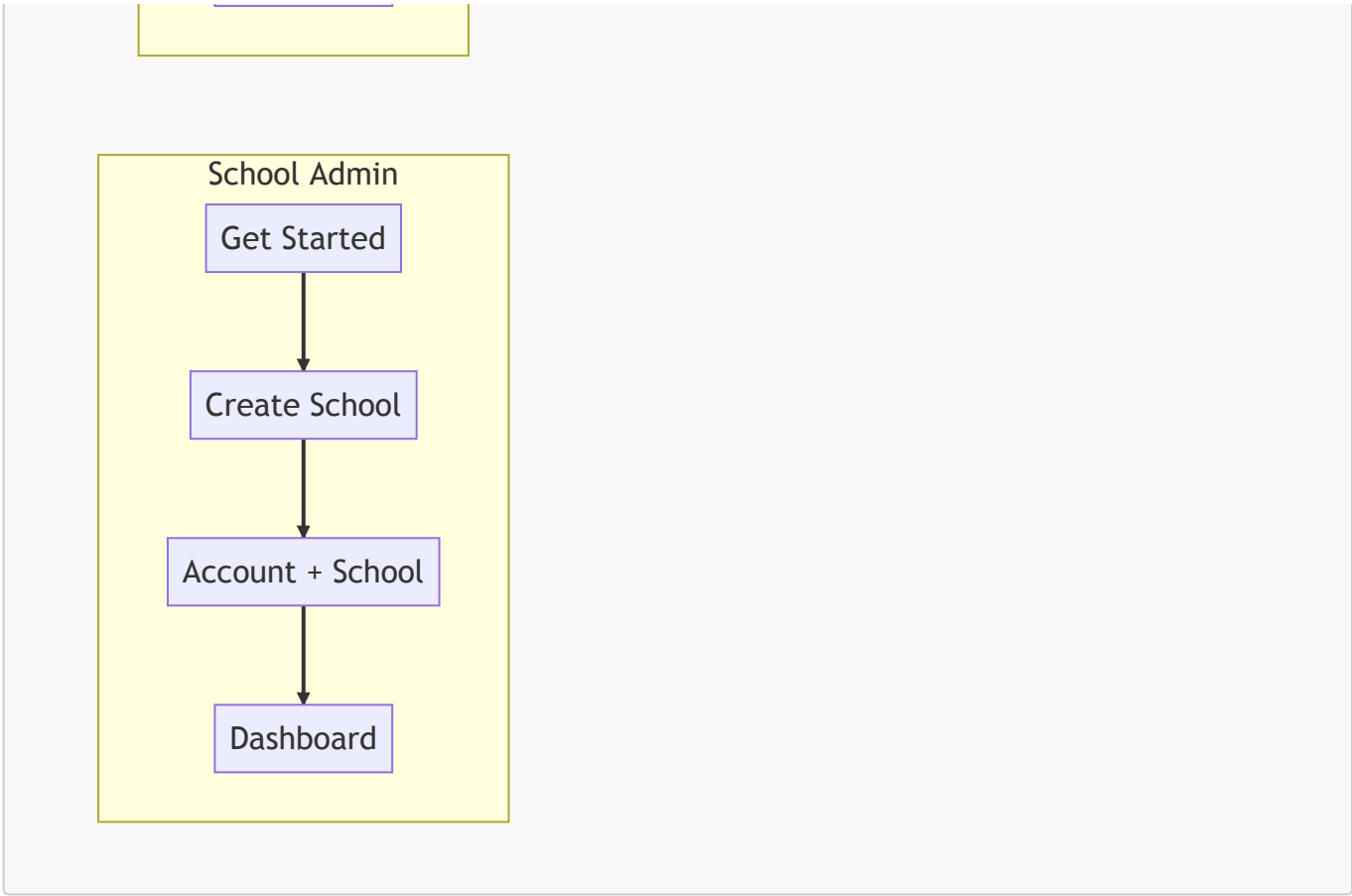
5.6 Onboarding Flow Diagram by Role




Parent Invite
Student enrolled
Parent auto-invited
Set password
Dashboard


Teacher Invite
Admin invites
Email with link
Set password
Dashboard


Existing User
Sign In
Credentials
Dashboard



5.7 Route Summary

Route	Purpose
`/`	Landing; "Get Started" → /create-school, "Sign In" → /sign-in
`/create-school`	New admin: create account + school
`/sign-in`	Existing users: authenticate
`/invite/[token]`	Invited users: accept invite, set password

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6. People Management

People management covers how admins add, edit, and remove teachers, students, and parents. It includes direct creation, the invite system, and bulk import.

6.1 List Pages and CRUD

Role	List Page	CRUD	Detail Page
Teachers	`/list/teachers`	Create, update, delete	`/list/teachers/[id]`
Students	`/list/students`	Create, update, delete	`/list/students/[id]`
Parents	`/list/parents`	Create, update, delete	`/list/parents/[id]`

- Search and filters (e.g. by class, grade, subject)
- Pagination

- Server Actions for all CRUD operations

6.2 Invite System

Admins add users by inviting them instead of creating accounts directly.

Feature	Implementation
Send invite	Admin enters email, role, optional metadata → system creates Invite, sends email with link
Invite status	PENDING, ACCEPTED, EXPIRED
Resend	Admin can resend pending invites
Accept flow	User clicks link → `/invite/[token]` → sets password → Auth + profile created, invite marked ACCEPTED (see Section 5.4 Invited User)
Bulk invite	CSV upload: email, role, name, … → create Invites, send emails
Invite on enrollment	When enrolling a student, optionally invite parent by email

6.3 Bulk Import

Entity	Method	Headers / Fields
Students	CSV upload	Username, password, name, grade, class, parent info, …
Teachers	CSV upload	Username, password, name, subjects, …
Results	CSV upload	Student, exam/assignment, score

- Validation and error reporting
- Creates Auth + profile records
- Optional: send "Set your password" emails for bulk-created users

6.4 Relationship to Onboarding

Flow	Section
Admin sends invite	People Management (this section)
User receives email, clicks link, sets password	User Onboarding (Section 5.4)
Admin creates user directly (no invite)	People Management
Admin bulk imports	People Management

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7. Role-Based Access Control

Access rules (from src/middleware.ts):

- `/schools/[schoolId]/*`: User must have `schoolId === schoolId` (no cross-school access)
- `/system/*`: Only `system_admin`
- Unauthenticated: Redirect to `/sign-in`

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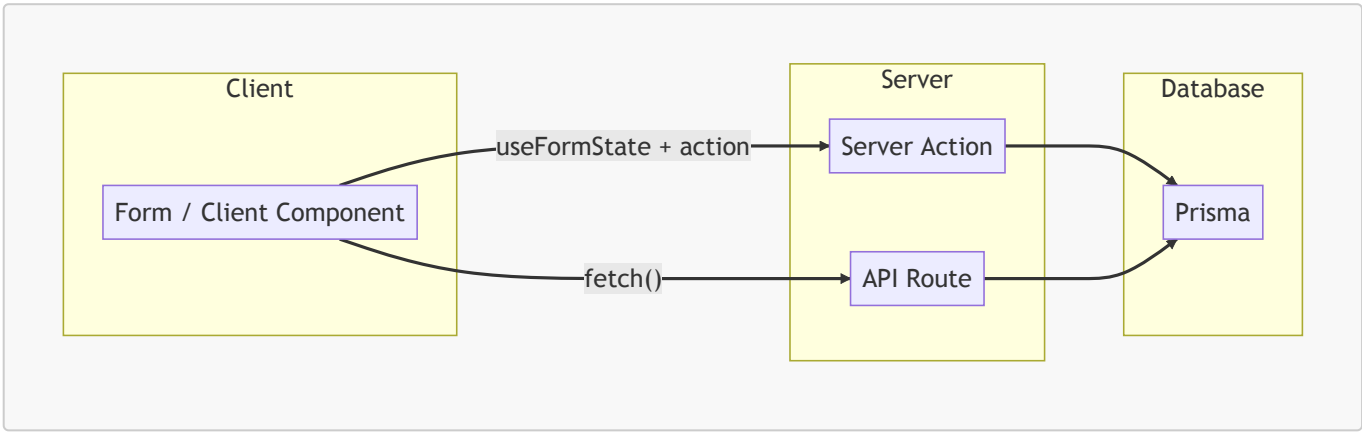
8. API Structure

Route	Methods	Purpose
`/api/auth/sign-in`	POST	Authenticate, set cookie
`/api/auth/sign-out`	GET	Clear cookie
`/api/auth/set-token`	POST	Set token (e.g. after sign-up)
`/api/auth/me`	GET	Current user (for Menu)
`/api/auth/sign-up`	POST	Register new user
`/api/schools/[schoolId]/academic-years`	GET, POST	Academic years CRUD
`/api/schools/[schoolId]/academic-years/[id]`	GET, PATCH, DELETE	Single academic year
`/api/schools/[schoolId]/academic-years/[id]/curriculum`	GET, POST	Curriculum
`/api/schools/[schoolId]/academic-years/[id]/classes/[classId]/enrollments`	GET, POST	Enrollments
`/api/schools/[schoolId]/rooms`	GET, POST	Rooms
`/api/schools/[schoolId]/rooms/[roomId]`	GET, PUT, DELETE	Single room
`/api/schools/[schoolId]/announcements`	GET	Announcements
`/api/schools/[schoolId]/subscriptions/subscribe`	POST	Stripe checkout
`/api/schools/[schoolId]/subscriptions/current`	GET	Current subscription
`/api/subscription-plans`	GET	Public plans
`/api/system_admin/subscription-plans`	GET, POST	Manage plans
`/api/system_admin/subscription-plans/[planId]`	GET, PUT, DELETE	Single plan
`/api/system_admin/school-subscriptions`	GET	All school subscriptions

Route	Methods	Purpose
`/api/system_admin/school-subscriptions/[id]`	GET, PUT	Single subscription
`/api/webhooks/payment`	POST	Stripe webhooks

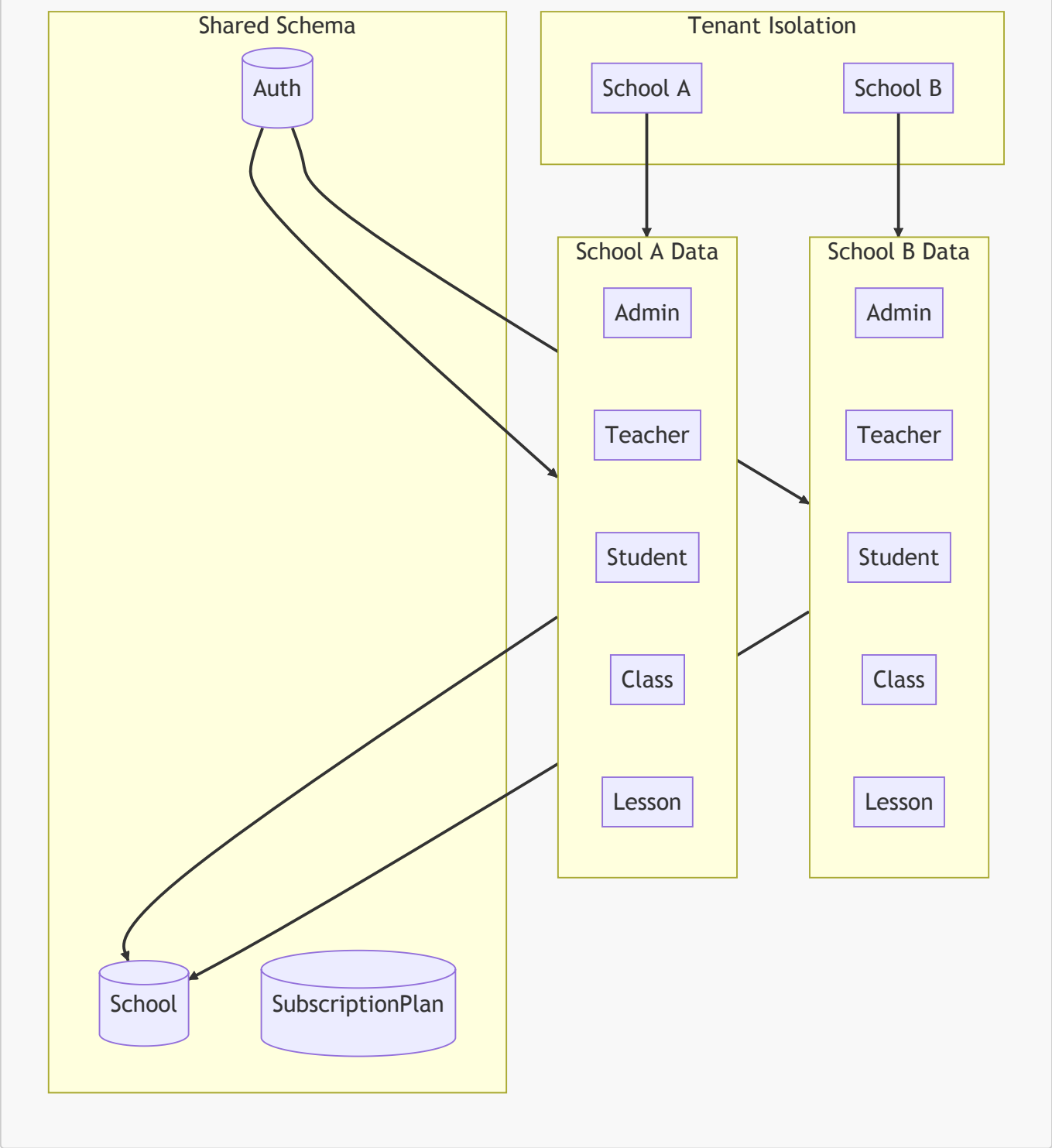
9. Data Flow: Server Actions vs API



Primary pattern: Server Actions in src/lib/actions.ts handle most CRUD (subjects, classes, grades, lessons, exams, assignments, attendance, rooms, schedule change requests, etc.). Actions use `getCurrentUserSchoolId()` / `getVerifiedAuthUser()` for auth and `revalidatePath()` for cache invalidation.

API routes are used for: auth, Stripe checkout, webhooks, and some read-only endpoints (announcements, subscription plans).

10. Multi-Tenant Architecture



Tenancy model: Row-level isolation via `schoolId` on all school-scoped tables. No separate DB per tenant. Middleware enforces `user.schoolId === path schoolId`.

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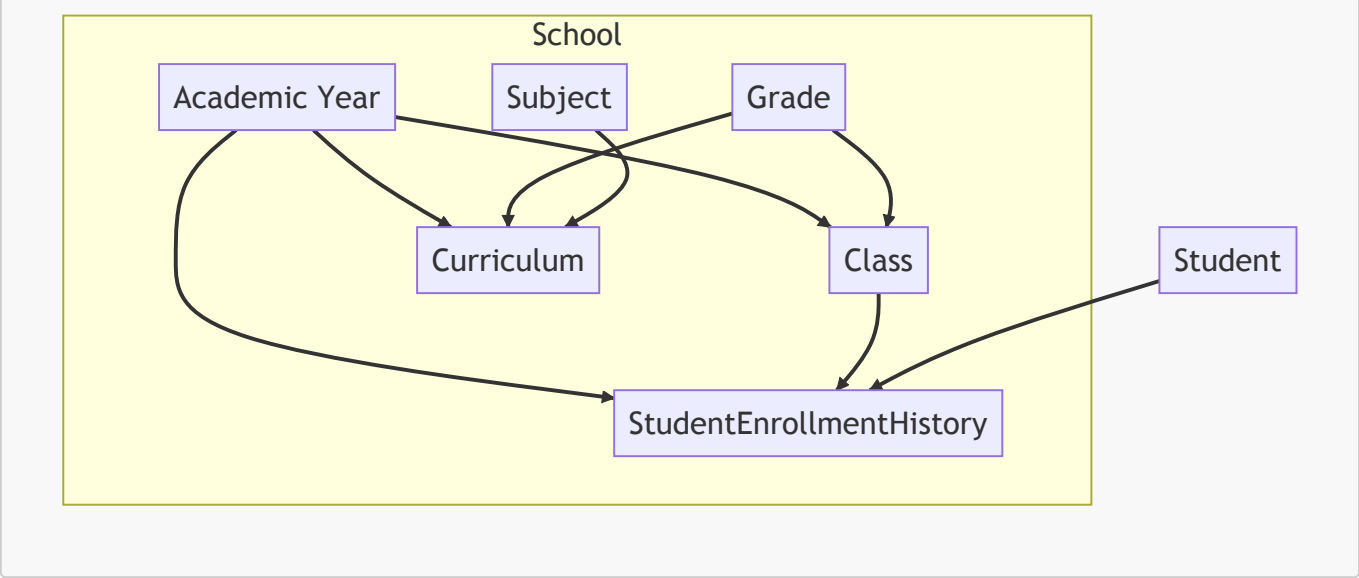
11. Academic Structure

11.1 Overview

The academic structure defines how schools organize grades, classes, subjects, curriculum, and student enrollment. It is the foundation for scheduling, lessons, exams, and attendance.



11.2 Core Entities

Entity	Purpose
AcademicYear	Bounds a school year; classes, curriculum, and enrollments are scoped to it
Grade	School-level grade level (e.g. Grade 1, Grade 5, Kindergarten)
Class	A specific section within a grade for an academic year (e.g. "5A", "Grade 3-B")
Subject	School-level subject (e.g. Math, French, Science)
Curriculum	Links an academic year to grades and subjects (what is taught in each grade)
StudentEnrollmentHistory	Records which class/grade a student is in for a given academic year

11.3 Enrollment as Source of Truth

Target design: `StudentEnrollmentHistory` is the canonical record of a student's class and grade for an academic year. `Student.classId` and `Student.gradeId` are derived and kept in sync.

Rule	Implementation
On enroll	Create `StudentEnrollmentHistory`; set `Student.classId`, `Student.gradeId` from the enrollment
On unenroll	Mark enrollment inactive; clear `Student.classId`, `Student.gradeId` if no other active enrollment
Display	Use enrollment data for "current class/grade"; fall back to `Student` fields for legacy or single-enrollment views

11.4 Curriculum Enforcement

Curriculum defines which subjects are taught in each grade. Lessons should reference curriculum entries to ensure consistency:

Rule	Implementation
Lesson creation	Lesson's subject should belong to the class's grade curriculum
Grade ordering	Grades have a `sortOrder` for display (K, 1, 2, …)
Capacity	Class has optional `capacity`; enrollment checks capacity before adding

11.5 Curriculum Templates by Country

School admins can optionally load a pre-built curriculum based on national education systems instead of creating from scratch.

11.5.1 Benefits

Benefit	Reason
Time savings	Schools avoid defining grades and subjects manually
Standards alignment	Matches national programmes (e.g. France, UK, US)
Consistency	Same labels mean similar content across schools
Onboarding	New schools can start quickly

11.5.2 Template Selection Flow

```
Admin → Academic Year → Curriculum setup

1. "How do you want to set up the curriculum?"
   - [Load from template] → Select country → Select system → Preview →
Apply
   - [Start from scratch] → Current behavior
   - [Copy from previous year] → If school has past curriculum

2. For template: Select country (e.g. "France")
3. System suggests system (e.g. "Éducation Nationale - Programmes 2024")
4. Preview: Grade 1 → French, Math, …; Grade 2 → …
5. Admin can add/remove subjects before applying
6. Apply → Creates Curriculum entries for the academic year
```

11.5.4 Initial Systems to Support

Country	System	Notes
France	Éducation Nationale	National programmes
USA	Common Core (Math, ELA)	State variations later

Country	System	Notes
UK	England National Curriculum	Scotland, Wales, NI later
Morocco	Programmes officiels	If relevant for target users
International	IB, Cambridge	For international schools

11.5.5 UX Summary

Question	Answer
Should schools get pre-built curricula?	Yes, as an optional template
By country?	Yes, country + system (e.g. "France - Éducation Nationale")
When?	During curriculum setup for an academic year
Required?	No – "Start from scratch" and "Copy from previous year" remain
Implementation	Start with a few systems, expand over time

12. Academic Activities

Academic activities are the day-to-day teaching and assessment operations built on top of the academic structure. All activities use enrollment as the source of truth for student–class relationships and respect academic year scope.

12.1 Lessons

Lessons are the core scheduling unit. Each lesson links a subject, class, teacher, room, day, and time slot.

Feature	Implementation
CRUD	Server Actions with conflict checks (teacher availability, teacher double-booking, class double-booking)
Working hours	8:00–17:00, weekdays only
Room assignment	Optional room with capacity checks
Curriculum enforcement	Lesson subject must belong to the class's grade curriculum
Academic year scope	Classes scoped by academic year when creating lessons
Teacher–subject validation	Teacher must teach the subject
List page	Search/filter by class, teacher, subject

12.2 Exams

Exams are formal assessments linked to lessons.

Feature	Implementation
CRUD	Server Actions
Lesson link	`lessonId` required; exams always tied to a lesson
Role-based filtering	Admin (all), teacher (own lessons), student (own class), parent (children)
List page	Subject, class, teacher, date; search by class, teacher, subject

12.3 Assignments

Assignments are homework or tasks linked to lessons.

Feature	Implementation
CRUD	Server Actions
Lesson link	`lessonId` required
Role-based filtering	Admin, teacher, student, parent
Student visibility	Based on enrollment history for the active academic year
List page	Title, subject, class, teacher, due date

12.4 Attendance

Attendance records student presence per lesson and date.

Feature	Implementation
CRUD	Server Actions
Teacher flow	Select lesson → take attendance for that lesson's class
Student list	From enrollment history (active enrollment for the lesson's class in the current academic year)
Status values	Present, Absent, Late
Form/schema alignment	Form sends `status`; schema expects `status`; update flow pre-fills from existing records
Admin overview	Recent attendance records
Student/parent views	Role-specific attendance pages
Charts	Attendance visualization components

12.5 Results

Results store scores for exams and assignments.

Feature	Implementation
CRUD	Server Actions
Link	Either exam or assignment (mutually exclusive)
Role-based filtering	Admin (all), teacher (own lessons), student (own), parent (children)
Null safety	Handles exams/assignments with or without lessons
Search	By student, exam/assignment title
Bulk import	Bulk create results

12.6 Curriculum

Curriculum defines which subjects are taught in each grade per academic year.

Feature	Implementation
CRUD	`curriculumActions.ts` (create, update, delete)
Scope	Academic year + grade
Unique constraint	One subject per grade per year
Lesson enforcement	Subject must be in curriculum for class's grade when creating lessons
Templates	Optional country-based templates (France, USA, UK, Morocco, IB/Cambridge)
UI	Curriculum page in academic year section

12.7 Data Flow Summary

Activity	Student Source	Curriculum Check	Academic Year Scope
Lessons	—	Yes	Yes (via class)
Exams	—	—	Yes (via lesson)
Assignments	Enrollment history	—	Yes (via lesson)
Attendance	Enrollment history	—	Yes (via lesson)
Results	—	—	Yes (via exam/assignment)
Curriculum	—	—	Yes

12.8 Enrollment as Source of Truth

`StudentEnrollmentHistory` is the canonical record of a student's class and grade for an academic year. `Student.classId` and `Student.gradeId` are kept in sync on enroll/unenroll. Academic activities (attendance, assignments, etc.) use enrollment data to determine which students belong to a class for a given academic year.

13. Scheduling

Scheduling manages lessons across the week, room allocation, teacher availability, and schedule change requests. All schedule views are scoped by academic year and use enrollment as the source of truth for student–class relationships.

13.1 Admin Schedule

Admins manage the school schedule via an interactive calendar.

Feature	Implementation
Calendar	FullCalendar (timeGridWeek, timeGridDay) with drag-and-drop and resize
Create lesson	Select time slot → form pre-filled with day, time, optional class/teacher from filters
Edit lesson	Click event → update form
Filters	Class filter; teacher filter with availability overlay
Teacher availability	Unavailable slots shown as red background when a teacher is selected
Room display	Room shown on events when assigned
Academic year scope	Lessons and classes filtered by active (or selected) academic year
Conflict checks	Teacher overlap, class overlap, room overlap, teacher availability
Events	School events (meetings, holidays) shown on the same calendar

13.2 Teacher Schedule

Teachers view their schedule and request changes.

Feature	Implementation
View	Calendar or list by day
Request change	"Request Change/Swap" per lesson → modal with TIME_CHANGE or SWAP
TIME_CHANGE	Proposed day, start time, end time
SWAP	Proposed swap teacher (filtered by subject)
Cancel	Teacher can cancel PENDING requests
Notifications	Notified when request is approved or rejected

13.3 Schedule Change Requests

Teachers request changes; admins approve or reject.

Feature	Implementation
Create	TIME_CHANGE or SWAP with validation
Admin page	Pending requests with approve/reject
Approve	Updates lesson (time or teacher) after conflict checks
Conflict checks on approve	Teacher overlap, class overlap, room overlap
Swap validation	Proposed teacher must teach the subject
Reject	Optional admin notes; teacher notified

13.4 Student Schedule

Students see their class schedule for the current academic year.

Feature	Implementation
Data source	Enrollment history for active academic year (not `Student.classId`)
Display	BigCalendar with lessons for enrolled class
Null handling	"No schedule" when not enrolled; no crash on missing class
Academic year	Only lessons for the active year

13.5 Parent Schedule

Parents see schedules for each enrolled child.

Feature	Implementation
Data source	Enrollment history per child for active academic year
Display	One schedule per child (tabs or stacked layout)
Null handling	Skips or shows "No schedule" for unenrolled children
Empty state	Handles parents with no children or no enrolled children

13.6 updateLessonTime (Drag/Resize)

When admins drag or resize lessons on the calendar:

Feature	Implementation
Updates	Day, startTime, endTime
Conflict checks	Teacher overlap, class overlap, room overlap, teacher availability
Revert	Reverts on conflict or error

13.7 Room Conflict Prevention

Room double-booking is prevented across all scheduling flows.

Flow	Check
createLesson	Room not already used at same day/time
updateLesson	Same as create
updateLessonTime	Same for new day/time
Approve TIME_CHANGE	Same for proposed day/time

13.8 Data Flow Summary

View	Lesson Source	Academic Year	Student Source
Admin schedule	All lessons (filtered by class/teacher)	Active year	—
Teacher schedule	Teacher's lessons	Active year	—
Student schedule	Class lessons	Active year	Enrollment
Parent schedule	Class lessons per child	Active year	Enrollment

13.9 Recurring Model

Lessons are weekly recurring: each lesson has `day`, `startTime`, `endTime` and repeats every week. `adjustScheduleToCurrentWeek` maps them to the current week for display. Events (meetings, holidays) are date-specific and shown alongside lessons on the admin calendar.

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14. Teacher Marketplace (Planned)

14.1 Overview

The Teacher Marketplace allows schools to discover and recruit teachers who are open to new opportunities. Teachers can work at multiple schools; the marketplace facilitates discovery, contact, and assignment.

Key concepts:

- **Multi-school teachers** – A teacher can have assignments at multiple schools
- **Marketplace profile** – Teachers opt in to be discoverable; schools browse and search
- **Contact flow** – Conversation → offer → agreement → assignment (not instant accept)

14.2 Multi-School Support

Teachers are linked to schools via a junction table instead of a direct `schoolId`.

- **Teacher** – One record per person (profile)
- **TeacherSchoolAssignment** – One record per school the teacher works at; lessons reference the assignment

14.3 Marketplace Profile

Teachers opt in to be discoverable. Profile includes structured availability (Option B) for time-saving matching:

Field	Purpose
headline	e.g. "Math & Physics Teacher, 5 years experience"
bio	Free-text description
subjectsOffered	Subjects they can teach
availabilityType	FULL_TIME, PART_TIME, SUBSTITUTE, TUTORING
availabilitySlots	Structured: `[{ day, startTime, endTime }, ...]` – schools see exact free slots
availabilityUpdatedAt	"Last updated" timestamp
isActive	Teacher can pause visibility

Availability display (Option B):

```
Availability for new work:
• Monday    09:00 – 12:00
• Wednesday 09:00 – 15:00
• Friday    14:00 – 17:00
Last updated: 3 days ago
```

14.4 Contact Flow (Not Instant Assign)

When a school contacts a teacher, the flow is conversational:

Stage	Description
CONTACTED	School sent initial message
IN_CONVERSATION	Messages exchanged
OFFER_SENT	School sent formal offer (subjects, dates, hours)
OFFER_NEGOTIATING	Teacher requested changes
AGREED	Both confirmed terms
ASSIGNED	TeacherSchoolAssignment created
DECLINED	Contact ended

14.6 Flow Summary

Actor	Action

Actor	Action
Teacher	Opts in, sets profile + structured availability slots
School	Browses marketplace, filters by subject/availability
School	Views teacher profile (sees availability slots)
School	Sends contact message
Both	Conversation (messages)
School	Sends formal offer (optional)
Teacher	Accepts/declines/negotiates offer
Both	Agree on terms
System	Creates TeacherSchoolAssignment

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15. Student Transfer (Planned)

15.1 Overview

Students can transfer from one school to another. The transfer uses a simple model: update `Student.schoolId` when the transfer is completed. Parents can have children at different schools; each student is at one school at a time.

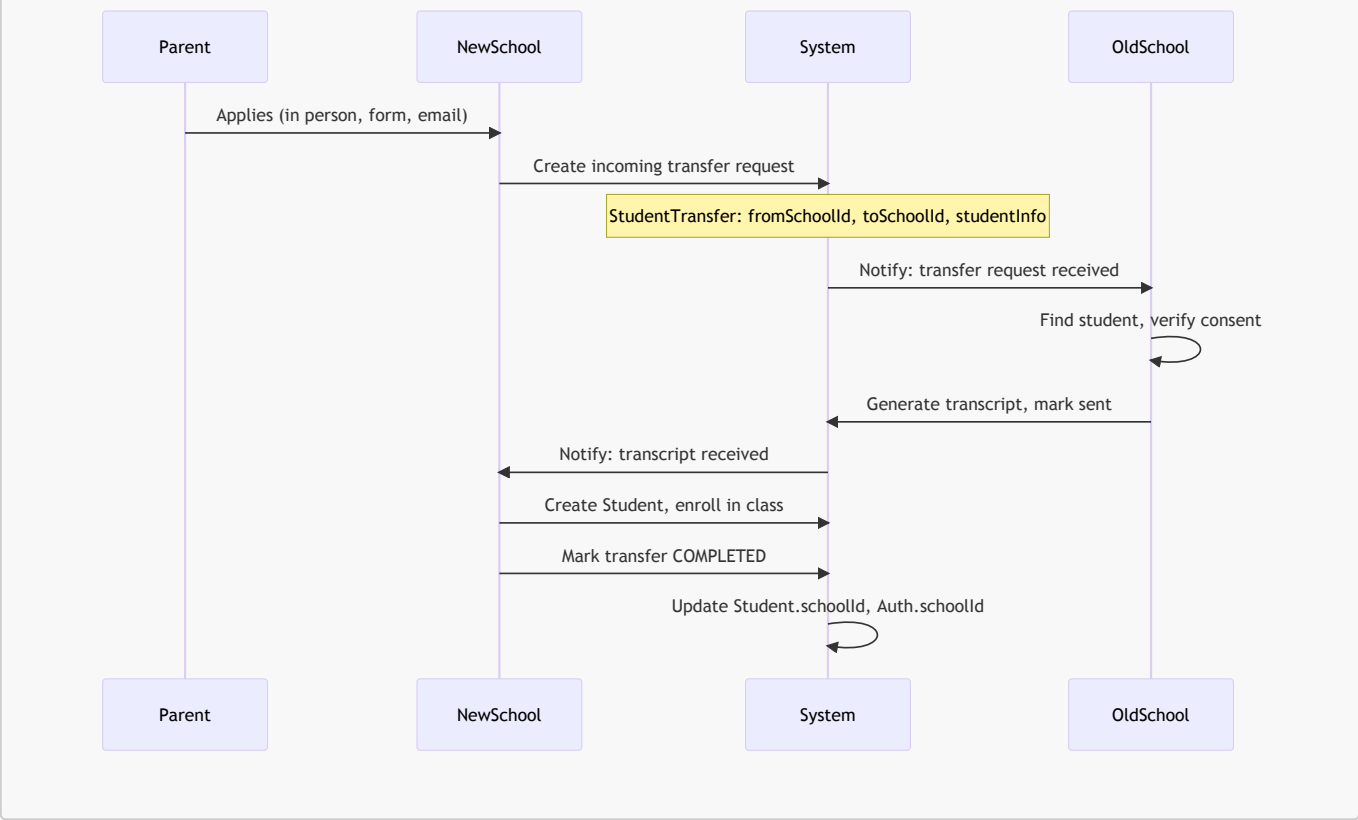
Key rules:

- **Student** – One school at a time; `schoolId` updated on transfer
- **Parent** – Can have children at different schools; access derived from children's schools
- **Transfer** – Request → transcript → enrollment → update `schoolId`

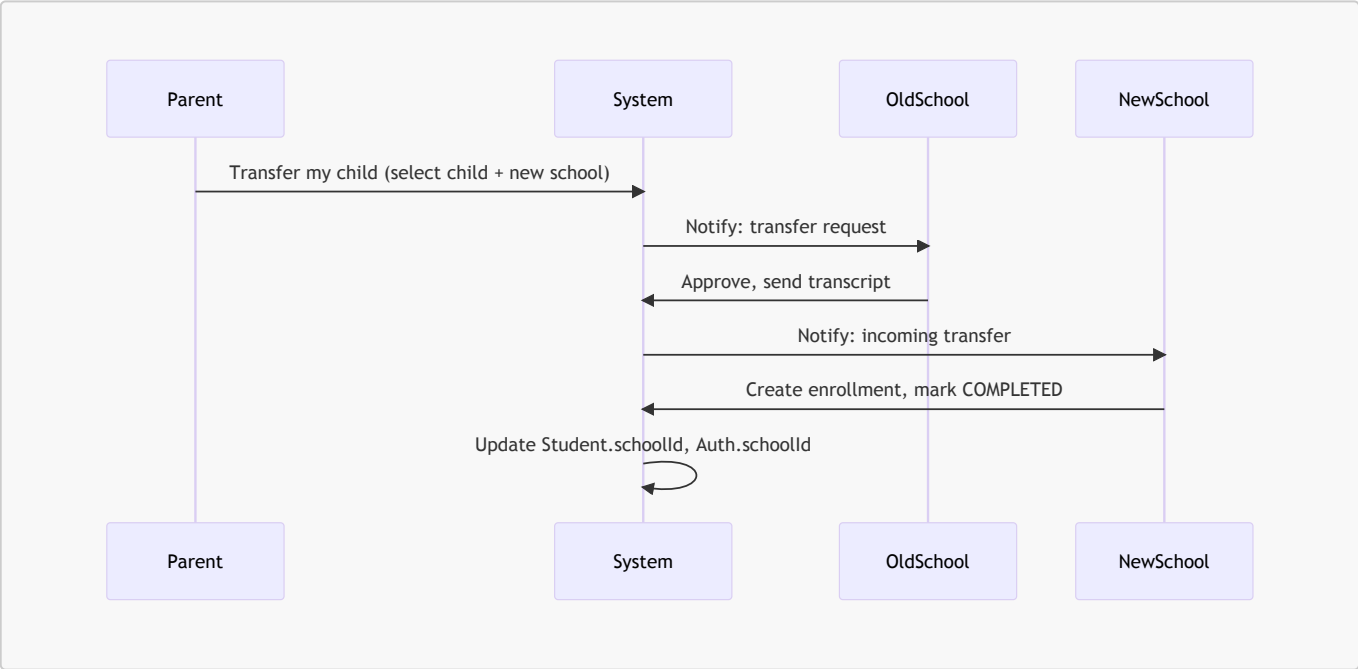
15.2 Who Initiates

Initiator	When	Flow
New school (primary)	Parent has applied to new school	New school creates incoming transfer request; old school sends transcript; new school enrolls
Parent	Both schools on Skooly, parent uses app	Parent requests transfer; old school approves and sends transcript; new school enrolls
Old school	Parent gave notice they're leaving	Old school creates transfer-out; prepares transcript for new school or parent

15.3 Transfer Flow (New School Initiates)



15.4 Transfer Flow (Parent Initiates)



15.5 Parent Multi-School Access

Parents derive school access from their children:

Rule	Implementation
Parent's schools	`students.map(s => s.schoolId).distinct`

Rule	Implementation
Access check	Parent can access `/schools/[schoolId]` if they have a Student with that `schoolId`
Schema	Remove or relax `Parent.schoolId`; access derived from children

When a child transfers, the parent gains access to the new school and may lose access to the old school if they have no other children there.

15.7 UI Placement

Initiator	Location
New school	Admin → Students → "Add incoming transfer" / "Enroll transferring student"
Parent	Parent dashboard → "Transfer my child" or student profile → "Request transfer"
Old school	Admin → Students → select student → "Transfer out"

15.8 Records Handling

Data	Handling
Old results, attendance	Stay at old school (historical)
Transcript	Generated by old school; sent to new school for placement
New enrollment	New school creates class/grade enrollment

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16. External Integrations

Service	Usage
Stripe	Subscription plans, checkout, webhooks (`checkout.session.completed`, `customer.subscription.*`)
Cloudinary (next-cloudinary)	Image uploads (profile, etc.)
PostgreSQL	Primary database via Prisma

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18. Implemented Features

The following parts of the app are already developed:

18.1 Core Platform

- Multi-tenant school management
- User roles (Admin, Teacher, Student, Parent, System Admin)
- JWT-based authentication (custom, HttpOnly cookie)

- Role-based access control and route protection
- Subscription management (Stripe integration)

18.2 People Management

- List pages and CRUD for teachers, students, parents
- Profile detail pages
- **Invite system** (send invite, accept flow at `/invite/[token]`, resend, bulk invite, invite on enrollment)
- Bulk import (students, teachers, results via CSV)

18.3 Academic Structure

- Academic year management
- Classes, Grades, Subjects
- Curriculum management (per academic year, per grade)
- Student enrollment history

18.4 Scheduling & Academic Activities

- Lessons and scheduling (FullCalendar, drag/drop, resize)
- Teacher availability management
- Schedule change requests (teacher-initiated time change/swap, admin approve/reject)
- Room/venue management
- Exams, Assignments, Results
- Attendance tracking

18.5 Other

- Events and Announcements
- Basic analytics (charts for attendance, counts, finance)

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19. Planned Features (from ECOSYSTEM_ROADMAP)

19.1 Core Missing Features

- Messaging & communication (WebSockets/SSE, direct and group messaging)
- Fee management and billing (student fees, payment tracking)
- Library management
- Transportation management
- Document management
- Advanced reporting and analytics
- Gradebook and transcript generation
- Notification system
- Parent-teacher conference scheduling

19.2 Cross-School Ecosystem

- **Teacher marketplace** (multi-school support, discovery, contact flow, structured availability)

- **Student transfer** (simple transfer, parent multi-school, new school / parent / old school initiation)
- Cross-school sports leagues & tournaments
- Alumni network across schools
- Group purchasing & bulk buying
- Shared transportation routes
- Cross-school parent networks
- Shared extracurricular activities